

FROM CIFAR CLASSIFIERS TO CLIP ENCODERS

# Invariance is not robustness

What actually predicts a model's adversarial robustness

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A model that ignores small shifts of its input is called invariant. A model that does not flip under a tiny worst-case change is called robust. We show these are different, measure what tells them apart, and correct a common mistake.

# The puzzle

Does making a vision model more shift-invariant help or hurt its adversarial robustness?

## One literature says: it helps.

Anti-aliasing and equivariant designs report better robustness. Smoother, more invariant models look safer.

## Another says: it hurts.

Perfect shift-invariance can shrink the margin between classes, which should make attacks easier.

Both can't be right as stated. Which is it, and what is the real signal?

**Takeaway.** We resolve this with one simple quantity, and find that the field's usual invariance metric is the wrong thing to look at.

# Two different things, often confused

## Invariance

Ignoring changes that should not matter:  
a small shift, a crop, a reworted prompt.



A shift is a change we *want* the model to ignore.

## Adversarial robustness

Not being flipped by a small *worst-case*  
change chosen by an attacker.

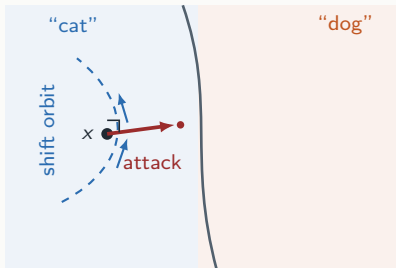


An attack is a change the model *must not* be  
fooled by.

They sound related, and the field often uses the first as a proxy for the second. Our first finding is that this proxy does not work.

# The picture: two different directions

Invariance moves *along* the boundary; an attack moves *across* it



- **Invariance** is being *flat along the shift orbit*: the dashed path stays inside the "cat" region, so the label never changes.
- **Robustness** is the *margin to the boundary*: the red arrow is the smallest change that crosses into "dog".
- These are **different directions**. A long, flat orbit (very invariant) can still sit a hair from the boundary (fragile).
- Invariance is a *projection*; robustness is a *distance*. One says little about the other.

**Takeaway.** Invariance and robustness live along near-orthogonal directions. This is the geometric reason a good invariance score does not buy you robustness.

# The measuring stick: a margin-to-sensitivity ratio

$\eta/L$ , computed on clean images, no attack needed

$$\eta/L = \frac{\text{margin}}{\text{sensitivity}} = \frac{\text{how confident the model is}}{\text{how fast that confidence changes if you nudge the input}}$$

- The **margin** is the gap between the top class and the runner-up.
- The **sensitivity** is the size of the input gradient of that margin, measured in the way the attacker moves (the norm matched to the threat).
- A large ratio means: far from the decision boundary, and slow to get there. That is exactly what robustness should be.

**Takeaway.**  $\eta/L$  is a first-order estimate of how big a perturbation it takes to flip the label. It needs no attack and is not changed by rescaling the model's outputs.

# Finding 1: the invariance metric is saturated

Shift-consistency looks the same for every modern model, robust or not

|                                                          | shift-consistency | spread across models |
|----------------------------------------------------------|-------------------|----------------------|
| 16 frozen CLIP encoders<br>(robust and non-robust alike) | 0.96 – 0.99       | almost none          |

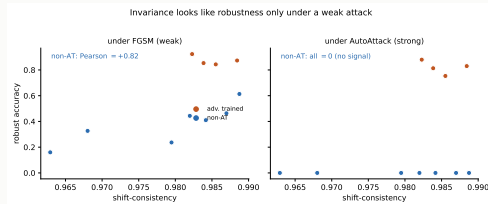
- Every modern encoder is already near-perfectly shift-consistent, so the metric cannot tell them apart on *any* kind of robustness.
- Once you account for clean accuracy, it does not predict robustness to common corruptions either.
- The same squeeze appears on strong CNN backbones: consistency stops discriminating.

**Takeaway.** Shift-consistency is treated as a robustness signal, but it is saturated. It has no room left to rank models. This is why the “invariance predicts robustness” story keeps giving mixed answers.

## Finding 2: “invariance helps” is a weak-attack artifact

The effect shows up under a cheap attack and vanishes under a strong one

- Under a single-step FGSM attack, more invariant models look more robust: the “invariance helps” ordering appears (left panel, a real trend).
- Under AutoAttack, every non-robust encoder drops to **zero**. There is no robustness left to order, and the trend disappears (right panel).
- A weak attack simply overstates robustness, the classic gradient-masking signature.



Same encoders, same budget: shift-consistency looks like it predicts robustness under FGSM, and predicts nothing under AutoAttack.

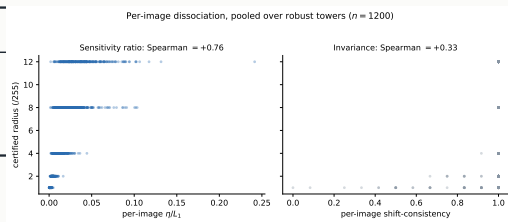
**Takeaway.** Recent claims that architectural invariance gives robustness without adversarial training are measured with weak attacks. They do not survive a strong one.

# Finding 3: the margin ratio predicts robustness

$\eta/L$  orders robustness where invariance does not, across three settings

| setting              | $\eta/L$   | consistency   |
|----------------------|------------|---------------|
| CIFAR-10 classifiers | +0.99      | uninformative |
| ResNet, ImageNet-100 | +0.90      | anti-corr.    |
| CLIP, per image      | +0.74–0.96 | $\approx 0$   |

- The same law from small CNNs to a ResNet to frozen foundation-model encoders.
- Inside one robust encoder, image by image,  $\eta/L$  tracks how large an attack each image needs; shift-consistency does not.



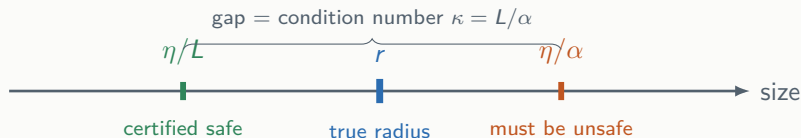
Per image, within a robust CLIP encoder:  $\eta/L$  (left) orders the attack size, shift-consistency (right) does not.

**Takeaway.** The right question is not “how invariant is the model” but “how large is the margin relative to the sensitivity”. That is  $\eta/L$ .



# Why it works: a two-sided bracket

The ratio is not just a heuristic; it brackets the true robust radius



- The **lower bound** is the standard certificate: the model cannot be flipped by anything smaller than  $\eta/L$ .
- The **upper bound** is the matching side: with  $\alpha$  a slope along the path to the boundary, no perturbation larger than  $\eta/\alpha$  can be safe.
- The gap between the two is a condition number  $\kappa = L/\alpha$ . When it is close to one,  $\eta/L$  is not just a bound, it is the answer.

**Takeaway.**  $\eta/L$  is the small-curvature limit of the true radius. This tells us exactly when it is tight, and points to what it misses.

## Finding 4: detecting robustness is not the same as ranking it

$\eta/L$  separates robust from fragile, but cannot order the robust ones

- Across all encoders,  $\eta/L$  cleanly **separates** the robust ones from the fragile ones.
- But *among the already-robust* encoders,  $\eta/L$  no longer orders them. Two encoders with the same  $\eta/L$  can differ in robustness.
- Something beyond the margin and the size of the gradient is at play.

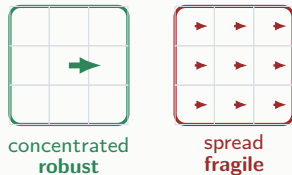
So what ranks the robust models?

**Takeaway.**  $\eta/L$  uses the *size* of the gradient. To rank robust models we need its *shape*.

## Finding 4: the shape of the gradient ranks robust models

A concentrated gradient resists an all-pixel attack better than a spread one

- An  $\ell_\infty$  attack pushes on *every* pixel at once.
- If the margin's gradient is **concentrated** on a few pixels, most of that budget is wasted, and the model holds.
- If it is **spread** over many pixels, the attack engages all of them, and the model gives way.
- Measured by the gradient's **anisotropy** (how spread it is), this orders the robust encoders where  $\eta/L$  could not.



An all-pixel attack wastes its budget on a concentrated gradient, but exploits every arrow of a spread one.

We measured this shape by the gradient's **anisotropy**  $A = \|\nabla M\|_1 / \|\nabla M\|_2$ : near 1 when concentrated,

up to  $\sqrt{d}$  when spread. It is the same  $\eta/L$  ingredients, read as a *ratio of norms* instead of a size.

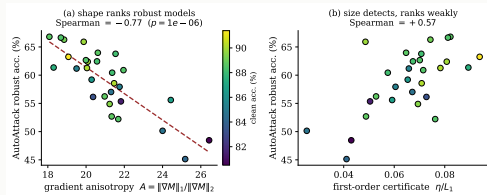
# Finding 4: why shape matters, and that it holds on real models

An effective-dimension deficit, not curvature, and it replicates on 30 CIFAR models

- $\eta/L_1$  is only the start. A real attack also harvests the gradient's variations across pixels, collecting *more* when the gradient is spread over more coordinates.
- This shrinks the true radius below  $\eta/L_1$  by a factor growing with the spread:

$$r \approx \frac{\eta/L_1}{1 + CA}$$

- A **transverse** effect, *not* curvature along the attack (on-direction curvature does **not** rank robustness). A matched-margin synthetic model reproduces the ordering exactly (rank corr.  $-1.0$ ).



30 RobustBench CIFAR-10 models. (a) the shape  $A$  ranks robustness (rank corr.  $-0.77$ , survives a clean-accuracy control); (b) the size  $\eta/L_1$  detects it but ranks weakly.

**Takeaway.** The shape law has a mechanism and replicates on 30 independent real robust models.

## Finding 5: adversarial training flips a hidden relationship

A fingerprint of what robust training actually changes

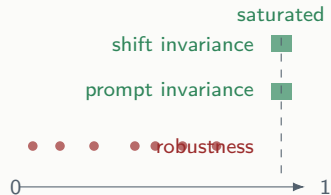
- In an ordinary model, the images with the largest margin tend to have the *smallest* gradients: they sit in flatter regions, so margin and robustness reinforce.
- After adversarial training, this reverses: high-margin images now have the *largest* gradients, so the model buys margin partly by raising sensitivity.
- The per-image relationship between margin and sensitivity flips sign, consistently across dozens of models.

**Takeaway.** Robust training does not just raise the margin. It re-couples margin and sensitivity, and  $\eta/L$  is the quantity that reads this change.

# Finding 6: the same story on images and text

On vision-language models, two kinds of invariance saturate, robustness does not

- A vision-language model sees an **image** and a **text prompt**. Both can be changed in ways that should not matter: shift the image, or reword the prompt.
- **Both invariances saturate**. Across many CLIP encoders, shift-consistency and prompt-paraphrase-consistency both sit near 0.98, robust or not. Neither can rank the models.
- **Adversarial robustness is a separate axis** with real spread, and the margin ratio  $\eta/L$  predicts it while neither invariance metric does.



Both invariances pin to the right edge;  
robustness is spread out.

**Takeaway.** Invariance is not robustness in *either* modality. The saturation of the invariance metrics carries from pixel shifts to prompts, and the margin ratio is what predicts robustness.

# What it all means

1. **Stop using shift-consistency as a robustness signal.** It is saturated on modern models and does not order adversarial robustness.
2. **Distrust “invariance helps” claims tested with weak attacks.** They vanish under AutoAttack.
3. **Use the margin-to-sensitivity ratio  $\eta/L$ .** It predicts robustness without running an attack, from CNNs to frozen CLIP encoders.
4. **To compare strong models, look at the gradient’s shape,** not only its size. Concentrated sensitivity is more robust.

One idea, the margin relative to the sensitivity, unifies a split literature and turns a vague intuition about invariance into something you can measure and predict.

# How to present this

The whole story in three beats



# The three-beat story

- Beat 1. The confusion.** People treat invariance and robustness as the same thing and get contradictory results. They are not the same thing.
- Beat 2. The fix.** A simple ratio, margin over sensitivity, predicts robustness without any attack, and the field's invariance metric does not, because it is saturated.
- Beat 3. The refinement.** That ratio tells robust from fragile; to rank the robust ones you also need the shape of the model's sensitivity, and the apparent “invariance helps” effect is just a weak-attack mirage.

**Takeaway.** If you remember one line: *invariance is not robustness; the margin relative to the sensitivity is.*

# Two analogies and the questions you will get

- **Analogy 1.** A locked door is not a strong door. Ignoring a gentle push (invariance) is not the same as withstanding a crowbar (an attack).
- **Analogy 2.** Two people equally far from a cliff edge: the one who moves slowly toward it is safer. That “distance over speed” is  $\eta/L$ .

**Q: Isn't the ratio just re-describing the radius it predicts?** Partly, and we say so; the surprise is that the invariance metric fails where the ratio succeeds.

**Q: Does it hold on big models?** Yes, on frozen CLIP encoders as well as CNNs.

**Q: Can you optimize it directly?** No, it is a diagnostic, not a training target; forcing it up collapses the margin.